Syllabus Details

Syllabus ID:	13245
Syllabus Name:	**Computer Organization and Architecture_Tổ chức và Kiến trúc máy tính**
Course Name English:	**Computer Organization and Architecture**
Subject Code:	**CEA201**
Learning-Teaching Method:	Social Constructivism, In-class lecture, Inquiry-based Teaching
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	Study hour (150h) = 45h contact hours + 1h final exam + 104h self-study
Pre-Requisite:	
Description:	This course in an introduction to computer architecture and organization. It will cover topics in both the physical design of the computer (organization) and the logical design of the computer (architecture). The main contents include the organization of a simple stored-program computer: CPU, busses and memory; Instruction sets, machine code, and assembly language; Conventions for assembly language generated by compilers; Floating-point number representation; Hardware organization of simple processors; Address translation and virtual memory; Very introductory examples of input/output devices, interrupt handling and multi-tasking systems. Chapter covered: Computer Evolution and Performance; A Top-Level View of Computer Function and Interconnection; Cache Memory; Internal Memory; External Memory; Input/Output; Operating System Support; Instruction Sets: Characteristics and Functions; Processor Structure and Function; Reduced Instruction Set Computers; Instruction-Level Parallelism and Superscalar Processors; Parallel Processing; Multicore Computers.
StudentTasks:	- Students must attend at least 80% of contact sessions in order to be accepted to the final examination. - Student is responsible to do all exercises and practical given by instructor in class or at home and submit on time. Do quizzes during class.
Tools:	- Intenet - Textbook - AI Tools and Simulator software
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	932/QĐ-ĐHFPT dated 08/22/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	8/22/2025

7 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	No
1	Computer Organization and Architecture: Design for Performance	William Stallings	Prentice Hall	2012	9th (or 10th)	978-0132936330 (9th), 978-0134101613 (10th)	True	True	False	
2	Computer architecture : A quantitative approach	John L. Hennessy, David A Patterson	Morgan Kaufmann				False	True	False	
3	http://williamstallings.com/ComputerOrganization/COA9e-Instructor/						False	False	True	

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	No
4	http://williamstallings.com/ComputerOrganization/COA9e-student/						False	False	True	
5	- AI Notes:https://www.https://notegpt.io/ or another AI. - Other AI tools: ChatGPT/Copilot/Gamma						False	False	True	
6	MARIE simulator						False	False	False	
7	ParaCacheSimulator						False	False	True	

10 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Explain the general functions and structure of a digital computer
2	CLO2	Present an overview of the evolution of computer technology from early digital computers to the latest microprocessors.
3	CLO3	Understand the key performance issues that relate to computer design
4	CLO4	Describe in detail the essential elements of computer organisation including internal bus, memory, Input/Output (I/O) organisations and interfacing standards and discuss how these elements function;
5	CLO5	Summarize, at a top level, the key functions of an operating system (OS).
6	CLO6	Be able to apply Boolean algebra and digital logic to design and interpret complex digital circuits;
7	CLO7	Present various types of addressing modes common in instruction sets, an essential characteristics of machine instructions, types of operands supported by typical machine instruction sets
8	CLO8	Explain processor structure and function in details, the operations of Reduced Instruction Set Computers
9	CLO9	Investigate, evaluate and communicate general trends in computing technologies such as Instruction-Level Parallelism and Superscalar Processors
10	CLO10	Using AI tools and simulator tools to futher explore the main concepts

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
1	- Introduction to the course. - Chapter 1: Introduction 1.1 Organization and Architecture	Offline	CLO1, CLO2, CLO10	IT	- Slide -Text Book - AI tool	CEA201	- Read chapter 1: text book, Slide - Use chatGPT(other suilable AI tool) to distinguish between organization and architecture using examples	
2	- Chapter 1: Introduction 1.2 Structure and Function	Offline	CLO1, CLO2, CLO10	T	- Slide -Text Book - AI tool	CEA201	- Read chapter 1: text book, slide - Use chatGPT(other suilable AI tool) to expore main keys: Von Neumann architecture (or Harvard architecture). - Use chatGPT to get examples to distinguish structure and function	
3	- Chapter 2: Computer Evolution and Performance 2.1 A Brief History of Computers	Offline	CLO1, CLO2, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 2 : text book, slide - Use chatGPT(other suilable AI tool) to expore the most modern generation of computers is associated with quantum computing, artificial intelligence (AI), and machine learning.	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
4	- Assessing exercises of chapter 1 - Chapter 2: Computer Evolution and Performance 2.2 Designing for Performance	Offline	CLO3, CLO10	TU	- Slide -Text Book - AI tool		- Read chapter 2 : text book, slide - Use ChatGPT to explore about "scalability and performance limits" - Presenting exercise chapter 1.	
5	- Chapter 2: Computer Evolution and Performance 2.3 Multicore, MICs, and GPGPUs	Offline	CLO3, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 2 : text book, slide - Use ChatGPT to futher explore about "Thread scheduling and CPU affinity","Distinguish among multicore, MIC, and GPGPU organizations". - Use chatGPT to find papers and research papers of NVIDIA	
6	- Chapter 2: Computer Evolution and Performance 2.6 Performance Assessment	Offline	CLO3, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 2: textbook, slide - Ask chatGPT to assess your computer capabilities	
7	- Assessing exercises of chapter 2 - Chapter 3: A Top-Level View of Computer Function and Interconnection 3.1 Computer Components	Offline	CLO4, CLO10	TU	- Slide -Text Book - AI tool		- Read chapter 3 : textbook, slide - Ask chatGPT to provide steps to build a virtual PC using a any PC building simulator or any activities to simulate the computer's components. - Presenting exercise chapter 2.	
8	- Chapter 3: A Top-Level View of Computer Function and Interconnection 3.2 Computer Function	Offline	CLO4, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 3: textbook, slide - Using a any PC building simulator to build a virtual PC or any activities to simulate the computer's components	
9	- Chapter 3: A Top-Level View of Computer Function and Interconnection 3.3 Interconnection Structures	Offline	CLO4, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 3: textbook, slide - Using a any PC building simulator to build a virtual PC or any activities to simulate the computer's components	
10	- Chapter 3: A Top-Level View of Computer Function and Interconnection 3.4 Bus Interconnection	Offline	CLO4, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 3: textbook, slide - Using a any PC building simulator to build a virtual PC or any activities to simulate the computer's components	
11	- Assessing exercises of chapter 3 '- Chapter 4: Cache Memory 4.1 Computer Memory System Overview	Offline	CLO4, CLO10	TU	- Slide -Text Book - AI tool		- Read chapter 4: textbook, slide - Ask chatGPT to get a Cache simulator to undestand about cache memory works - Ask chatGPT about "how to use a ParaCache simulator" or any other suilable simulator - Presenting exercise chapter 3.	
12	- Chapter 4: Cache Memory 4.2 Cache Memory Principles	Offline	CLO4, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 4: textbook, slide - Ask chatGPT to get sample data/program to run simulator (cache size, block-size, mapping methods: direct)	
13	- Chapter 4: Cache Memory 4.3 Elements of Cache Design	Offline	CLO4, CLO10	T	- Slide -Text Book - AI tool		- Modify parameters (cache-size, block-size, use mapping method: associativity) and run simulator again - ask chatGPT "Why did increasing associativity improve the hit rate?" or "What if I double the block size?"	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
14	- Chapter 4: Cache Memory 4.3 Elements of Cache Design (cont.)	Offline	CLO4, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 4: textbook, slide	
15	- Chapter 5: Internal Memory 5.1 Semiconductor Main Memory	Offline	CLO4,CLO10	T	- Slide -Text Book - AI tool		- Read chapter 5:textbook, slide - Use a MARIE simulator/any suilable simulator to explore the computer's registers	
16	- Assessing exercises of chapter 4 '- Chapter 5: Internal Memory 5.2 Error Correction	Offline	CLO4,CLO10	TU	- Slide -Text Book - AI tool		- Ask chatGPT to get some MARIE programs or any suilable programs to run simulator - Presenting exercises of chapter 4	
17	- Chapter 5: Internal Memory 5.3 Advanced Dram Organization	Offline	CLO4, CLO5, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 5: textbook, slide - Ask chatGPT to show a visual of how data flows from RAM to cache to CPU. - Ask chatGPTto explain this memory layout diagram step-by-step.	
18	- Chapter 5: Internal Memory 5.3 Advanced Dram Organization (cont)	Offline	CLO4, CLO5, CLO10	T	- Slide -Text Book - AI tool		- Ask chatGPT to generate a sample C program - Run it in the debug mode(use any IDE) to watch RAM in real program.	
19	- Assessing exercises of chapter 5 - Chapter 6: External Memory 6.1 Magnetic Disk	Offline	CLO4, CLO5, CLO10	TU	- Slide -Text Book - AI tool		- Read chapter 6: textbook, slide - Use chatGPT to futher explore about:Compare HDD and SSD for general users vs data centers. - Presenting exercises of chapter 5	
20	- Chapter 6: External Memory 6.2 Raid	Offline	CLO4, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 6: textbook, slide - Ask chatGPTto simulate what happens step-by-step when you open a file stored on an HDD. - Ask chatGPT “What are the performance limits of a 7200 RPM drive?”	
21	- Chapter 6: External Memory 6.2 Raid (cont)	Offline	CLO4	T	- Slide -Text Book - AI tool		- Ask chatGPT to generate a MARIE script to simulate disk scheduling algorithms. - Ask chatGPT to summary main keys about RAID - Ask chatGPT to generate quize about computer's memory	
22	- Chapter 6: External Memory 6.3 Solid State Drives	Offline	CLO4	T	- Slide -Text Book - AI tool		- Read chapter 6: textbook, slide - Ask chatGPT to compare I/O mapped I/O vs memory mapped I/O with examples	
23	- Chapter 7: Input/Output 7.1 External Devices 7.2 I/O Modules	Offline	CLO4	T	- Slide -Text Book - AI tool		- Read chapter 7: textbook, slide '- Ask chatGPT to go deeper: how a DMA controller works with a memory diagram.	
24	- Chapter 7 Input/Output 7.3 Programmed I/O	Offline	CLO4	T	- Slide -Text Book - AI tool		- Read chapter 7: textbook, slide '- Ask chatGPT to generate a labeled diagram showing the flow of data using DMA.	
25	- Assessing exercises of chapter 6 - Chapter 7 Input/Output 7.4 Interrupt-Driven I/O	Offline	CLO4, CLO10	TU	- Slide -Text Book - AI tool		- Read chapter 7: textbook, slide - Presenting exercises of chapter 6	
26	- Chapter 7: Input/Output 7.6 I/O Channels and Processors	Offline	CLO4, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 7: textbook, slide	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
27	Chapter 8: Operating System Support 8.1 Operating System Overview	Offline	CLO5, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 8: textbook, slide	
28	- Assessing exercises of chapter 7 - Chapter 8: Operating System Support 8.1 Operating System Overview (cont)	Offline	CLO5, CLO10	TU	- Slide -Text Book - AI tool		- Read chapter 8: textbook, slide - Presenting exercises of chapter 7	
29	- Chapter 8: Operating System Support 8.2 Scheduling	Offline	CLO5, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 8: textbook, slide '- Use ChatGPT to learn about topics related to FCFS, Shortest Job First, Round Robin, and Priority.	
30	- Chapter 8: Operating System Support 8.3 Memory Management	Offline	CLO5, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 8: textbook, slide - Use AI Notes/other suilable AI Q&A to answer questions related to memory management in the operating system	
31	- Assessing exercises of chapter 8 - Chapter 11: Digital Logic 11.1 Boolean Algebra 11.2 Gates	Offline	CLO5, CLO10	TU	- Slide -Text Book - AI tool		- Read chapter 11: textbook, slide - Ask chatGPT to explain AND, OR, NOT gates with real-life examples. - Presenting exercises of chapter 8	
32	- Chapter 11: Digital Logic 11.3 Combinational Circuit	Offline	CLO6, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 11: textbook, slide - Ask chatGPT to create a visual of a simple logic gate circuit. - Ask chatGPT to explain step by step to create a logic gate circuit.	
33	- Chapter 12: Instruction Sets: Characteristics and Functions 12.1 Machine Instruction Characteristics	Offline	CLO6, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 12 :textbook, slide - Use chatGPT to futher explore about: main keys	
34	- Assessing exercises of chapter 11 - Chapter 12: Instruction Sets: Characteristics and Functions 12.2 Types of Operands	Offline	CLO7,CLO10	TU	- Slide -Text Book - AI tool		- Read chapter 12 :textbook, slide - Use chatGPT to futher explore about: main keys. '- Presenting exercises of chapter 11	
35	- Chapter 12: Instruction Sets: Characteristics and Functions cont. 12.4 Types of Operations	Offline	CLO7,CLO10	T	- Slide -Text Book - AI tool		- Read chapter 12 :textbook, slide - Use chatGPT to futher explore about: main keys	
36	- Chapter 12: Instruction Sets: Characteristics and Functions cont. 12.4 Types of Operations	Offline	CLO7,CLO10	T	- Slide -Text Book - AI tool		'- Read chapter 12 :textbook, slide	
37	- Assessing exercises of chapter 12 - Chapter 13: Instruction Sets: Addressing Modes and Formats 13.1 Addressing Modes	Offline	CLO7, CLO10	TU	- Slide -Text Book - AI tool		- Read chapter 13 :textbook, slide '- Presenting exercises of chapter 12	
38	- Chapter 13: Instruction Sets: Addressing Modes and Formats 13.3 Instruction Formats 13.5 Assembly Language	Offline	CLO7,CLO10	T	- Slide -Text Book - AI tool		- Read chapter 13 :textbook, slide - Ask chatGPT to generate some assemly program to run MARIE/ other simulator	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
39	Practical Assembly Laguage	Offline	CLO7,CLO10	U	- Slide -Text Book - AI tool		Do exercises	
40	Practical Assembly Laguage	Offline	CLO7,CLO10	U	- Slide -Text Book - AI tool		Do exercises	
41	Practical Assembly Laguage	Offline	CLO7,CLO10	U	- Slide -Text Book - AI tool		Do exercises	
42	Practical Assembly Laguage	Offline	CLO7,CLO10	U	- Slide -Text Book - AI tool		Do exercises	
43	Practical Assembly Laguage	Offline	CLO7,CLO10	U	- Slide -Text Book - AI tool		Do exercises	
44	Practical Assembly Laguage	Offline	CLO7,CLO10	U	- Slide -Text Book - AI tool		Do exercises	
45	Assignments introduction	Offline	All CLOs	TU	- Slide -Text Book - AI tool		Do exercises	
46	- Assessing exercises of chapter 13 - Chapter 14: Processor Structure and Function 14.1 Processor Organization	Offline	CLO8, CLO10	TU	- Slide -Text Book - AI tool		- Read chapter 14: textbook, slide - Use chatGPT to futher explore about: main keys - Presenting exercises of chapter 13	
47	- Chapter 14: Processor Structure and Function 14.2 Register Organization	Offline	CLO8, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 14: textbook, slide - Use chatGPT to futher explore about: main keys	
48	- Chapter 14: Processor Structure and Function 14.3 Instruction Cycle	Offline	CLO9, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 14: textbook, slide - Use chatGPT to futher explore about: main keys	
49	- Chapter 14: Processor Structure and Function 14.4 Instruction Pipelining	Offline	CLO9, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 14: textbook, slide - ask chatGPT to create a MARIE script/other language script to simualte pipelining technology.	
50	- Chapter 15: Reduced Instruction Set Computers 15.1 Instruction Execution Characteristics 15.2 The Use of a Large Register File	Offline	CLO9, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 15: textbook, slide - Use chatGPT to futher explore about: main keys	
51	- Chapter 15: Reduced Instruction Set Computers 15.3 Compiler-Based Register Optimization 15.4 Reduced Instruction Set Architecture	Offline	CLO9, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 15: textbook, slide - Use AI Note to create quize and student answer. - Use AI Notes to get explaination	
52	- Assessing exercises of chapter 14 & 15	Offline	CLO9, CLO10	TU	- Slide -Text Book - AI tool		- Presenting exercises of chapter 14 & 15	

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Student's Tasks	URLs
53	- Chapter 16: Instruction-Level Parallelism and Superscalar Processors 16.1 Overview 16.2 Design Issues	Offline	CLO9, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 16: textbook, slide - Use chatGPT to futher explore about: main keys	
54	- Chapter 17: Parallel Processing 17.1 Multiple Processor Organizations 17.2 Symmetric Multiprocessors	Offline	CLO9, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 17: textbook, slide - Use chatGPT to futher explore about: main keys	
55	- Assessing exercises of chapter 16 - Chapter 17: Parallel Processing 17.3 Cache Coherence and the MESI Protocol 17.4 Multithreading and Chip Multiprocessors	Offline	CLO9, CLO10	TU	- Slide -Text Book - AI tool		- Read chapter 17: textbook, slide - Use chatGPT to futher explore about: main keys - Assessing exercises of chapter 16	
56	- Chapter 18: Multicore Computers 18.1 Hardware Performance Issues 18.2 Software Performance Issues	Offline	CLO9, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 18: textbook, slide - Use chatGPT to futher explore about: main keys	
57	- Chapter 18: Multicore Computers 18.3 Multicore Organization	Offline	CLO9, CLO10	T	- Slide -Text Book - AI tool		- Read chapter 18: textbook, slide - Use chatGPT to futher explore about: main keys	
58	- Assessing exercises of chapter 17 & 18	Offline	CLO9, CLO10	TU	- Slide -Text Book - AI tool		- Presenting exercises of chapter 17 & 18	
59	- Assessing assigments Review 1	Offline	All CLOs	U				
60	- Assessing assigments Review 2	Offline	All CLOs	U				

79 Constructive question(s)

No.	Session No	Name	Details
1	1	CQ1.1	What is computer architecture? What is computer organization? Example?
2	1	CQ1.2	What is the distinction between computer organization and computer architecture?
3	2	CQ1.3	Explain about Structure of computer. Explain about Functions of computer.
4	2	CQ1.4	What is the distinction between computer structure and computer function?
5	3	CQ1.5	Present an overview of the evolution of computer through generations.
6	4	CQ2.1	List and briefly define some of the techniques used in contemporary processors to increase speed.
7	4	CQ2.2	Explain the concept of performance balance.
8	5	CQ2.3	Distinguish among multicore, MIC, and GPGPU organizations.
9	6	CQ2.4	Explain about System Clock.
10	6	CQ2.5	Summarize some of the issues in computer performance assessment.
11	7	CQ3.1	Explain about major components of computer in a Top-lever view.
12	8	CQ3.2	What general categories of functions are specified by computer instructions?

No.	Session No	Name	Details
13	8	CQ3.3	List and briefly define the possible states that define an instruction execution?
14	8	CQ3.4	List and briefly define two approaches to dealing with multiple interrupts?
15	9	CQ3.5	What types of transfers must a computer's interconnection structure (e.g., bus) support?
16	10	CQ4.1	What is the benefit of using a multiple-bus architecture compared to a single-bus architecture?
17	11	CQ4.2	What are the differences among sequential access, direct access, and random access?
18	11	CQ4.3	What is the general relationship among access time, memory cost, and capacity?
19	12	CQ4.4	What is the cache memory?
20	12	CQ4.5	How does the principle of locality relate to the use of multiple memory levels?
21	13	CQ5.1	Explain about key elements of cache design.
22	13	CQ5.2	Distinguish among direct mapping, associative mapping, and set-associative mapping.
23	14	CQ5.3	Explain the reasons for using multiple levels of cache. Explain the performance implications of multiple levels of memory.
24	15	CQ5.4	What are two interpretations of the term random-access memory. What is the difference between DRAM and SRAM in terms of application. What is the difference between DRAM and SRAM in terms of characteristics such as speed, size, and cost. Explain why one type of RAM is considered to be analog and the other digital?
25	15	CQ5.5	What are some applications for ROM? What are the differences among EPROM, EEPROM, and flash memory?
26	16	CQ6.1	Explain about error incident in a semiconductor memory system and error Correction.
27	16	CQ6.2	How is the syndrome for the Hamming code interpreted.
28	17	CQ6.3	How does SDRAM differ from ordinary DRAM?
29	18	CQ6.4	What is the DDR DRAM?
30	19	CQ7.1	What are the advantages of using a glass substrate for a magnetic disk? How are data written onto a magnetic disk? How are data read from a magnetic disk?
31	19	CQ7.2	Define the terms seek time, rotational delay, access time, and transfer time in disk performance parameters?
32	20	CQ7.3	What common characteristics are shared by all RAID levels?
33	20	CQ7.4	Briefly define the seven RAID levels.
34	21	CQ7.5	How is redundancy achieved in a RAID system. In the context of RAID, what is the distinction between parallel access and independent access?
35	22	CQ8.1	Compare and contrast hard disk drives and solid disk drives.
36	23	CQ8.2	List three broad classifications of external, or peripheral, devices.
37	23	CQ8.3	What is the International Reference Alphabet (IRA)? Give an example. What are the major functions of an I/O module?
38	24	CQ8.4	List and briefly define three techniques for performing I/O?
39	24	CQ8.5	What is the difference between memory-mapped I/O and isolated I/O?
40	25	CQ9.1	What is the difference between programmed I/O and interrupt-driven I/O and discuss their relative merits.
41	25	CQ9.2	When a device interrupt occurs, how does the processor determine which device issued the interrupt. When a DMA module takes control of a bus, and while it retains control of the bus, what does the processor do?
42	26	CQ9.3	Present the evolution of the I/O Function.
43	26	CQ9.4	Explain about characteristics of I/O Channels.
44	27	CQ9.5	Summarize, at a top level, the key functions of an operating system (OS)
45	28	CQ10.1	Discuss the evolution of operating systems for early simple batch systems to modern complex systems.
46	29	CQ10.2	What is the difference between a process and a program?
47	29	CQ10.3	List and briefly define the major types of OS scheduling.
48	30	CQ10.4	Present some solutions in memory management. Assess the relative advantages of paging and segmentation.
49	30	CQ10.5	What is the virtual memory? What is the purpose of a translation lookaside buffer?
50	31	CQ11.1	What are digital circuits? What are the basic electronic components?

No.	Session No	Name	Details
51	32	CQ11.2	How can minimize a combinational circuits? Explain two methods and give examples.
52	33	CQ11.3	What is a machine instruction? What are the typical elements of a machine instruction?
53	34	CQ12.1	List and briefly explain five important instruction set design issues.
54	35	CQ12.2	Why are transfer of control instructions needed? List and briefly explain two common ways of generating the condition to be tested in a conditional branch instruction.
55	36	CQ12.3	What is meant by the term nesting of procedures? List three possible places for storing the return address for a procedure return.
56	37	CQ13.1	Describe the various types of addressing modes common in instruction sets. Present an overview of x86 and ARM addressing modes.
57	38	CQ13.2	Summarize the issues and trade-offs involved in designing an instruction format. Present an overview of x86 and ARM instruction formats.
58	39	CQ13.3	Distinguish between machine language and computer language. Explain and give some examples to illustrate the differences between them.
59	40	CQ14.1	List some reasons why it is worthwhile to study assembly language programming.
60	41	CQ14.2	What is an assembly language? List some advantages/disadvantages of assembly language compared to high-level languages.
61	42	CQ14.3	What are the typical elements of an assembly language statement?
62	43	CQ15.1	List and briefly define four different kinds of assembly language statements.
63	44	CQ15.2	What is the difference between a one-pass assembler and a two-pass assembler?
64	45	CQ15.3	How is a program compiled?
65	46	CQ16.1	Develop some basic console applications.
66	47	CQ16.2	What general roles are performed by processor registers? What categories of data are commonly supported by user-visible registers?
67	48	CQ16.3	What is the function of condition codes? What is a program status word?
68	49	CQ17.1	What is a program status word? Why is a two-stage instruction pipeline unlikely to cut the instruction cycle time in half, compared with the use of no pipeline?
69	50	CQ17.2	List and briefly explain various ways in which an instruction pipeline can deal with conditional branch instructions. How are history bits used for branch prediction?
70	51	CQ17.3	What are some typical distinguishing characteristics of RISC organization? Briefly explain the two basic approaches used to minimize register-memory operations on RISC machines.
71	52	CQ18.1	If a circular register buffer is used to handle local variables for nested procedures, describe two approaches for handling global variables. What are some typical characteristics of a RISC instruction set architecture? What is a delayed branch?
72	53	CQ18.2	What is the essential characteristic of the superscalar approach to processor design? What is the difference between the superscalar and super pipelined approaches?
73	54	CQ18.3	What is instruction-level parallelism? What is the distinction between instruction-level parallelism and machine parallelism?
74	55	CQ19.1	List and briefly define three types of computer system organization. What are the chief characteristics of an SMP (symmetric multiprocessor)? What are some of the potential advantages of an SMP compared with a uniprocessor? What are some of the key OS design issues for an SMP?
75	56	CQ19.2	What is the difference between software and hardware cache coherent schemes? What is the meaning of each of the four states in the MESI protocol?
76	57	CQ19.3	What are differences among simple instruction pipelining, superscalar, and simultaneous multithreading.
77	58	CQ20.1	Give several reasons for the choice by designers to move to a multicore organization rather than increase parallelism within a single processor.
78	59	CQ20.2	Why is there a trend toward giving an increasing fraction of chip area to cache memory?
79	60	CQ20.3	At a top level, what are the main design variables in a multicore organization? List some advantages of a shared L2 cache among cores compared to separate dedicated L2 caches for each core.

3 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assignment	on-going	2	20.0%	>0	At home	All CLOs	Option 1: Write a report, or developing assemly program Option 2: Follow lecturer's proposal	Option 1: build a simple vitrual PC/other suilable activities and a report. Or Develop assemly program Option 2: Follow lecturer's proposal	Basic computer components/Basic programs	Teachers assess their works on report	
2	Exercices	on-going	4	40.0%	>0	Do at home Present in class	All CLOs	Apply AI to answer and explore the content of the subject based on the lecturer's questions and submit the result to Edunext	Follow lecturer's proposal on chapters	Studied chapters; knowledge and skills	Teachers assess their works on report	
3	Final exam	Final exam	1	40.0%	4	60'		Multiple choice	50	All chapters; knowledge and skills of digital system;	by Exam board	